

RAK19012 WisBlock USB LiPo Solar Power Slot Module Datasheet

Overview

Description

The RAK19012 WisBlock USB LiPo Solar Power Slot Module is a power board comprising a USB-C connector, battery connector with an onboard charger, solar panel connector, LED indicator for charge status, two user-configurable LEDs, a reset button, and a power connector for connection with the WisBlock Base board. This power board allows debugging and firmware uploading to the WisBlock Core via the USB-C connector.

Features

- USB-C connector for programming and debugging the WisBlock Core
- Compatible with LiPo rechargeable batteries
- Solar panel connector for battery charging
- Onboard battery charger chip
- LED for charging status and user-configurable LEDs
- Module size: 30 mm X 20 mm

Specifications

Overview

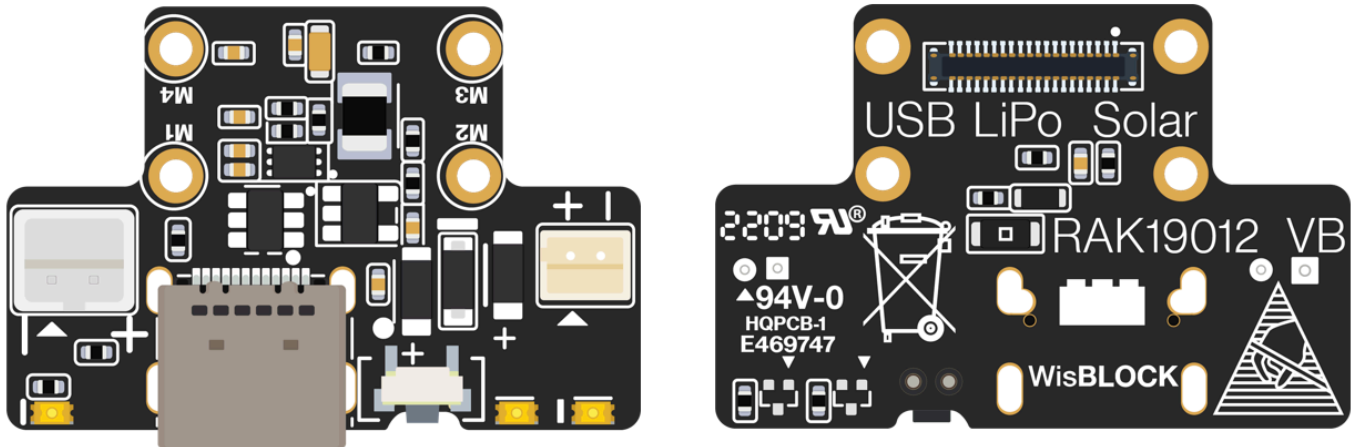


Figure 1: RAK19012 WisBlock Power Module top (left) and bottom (right) view

Mounting Sketch

The RAK19012 module can be mounted on the power slot of a WisBlock Base board with a power slot. **Figure 2** shows the mounting mechanism of the RAK19012 on a WisBlock Base module, such as the RAK19010.

WARNING

The RAK19012 **only** supports WisBlock Base boards with a power slot. It is not compatible with all WisBlock Base boards.

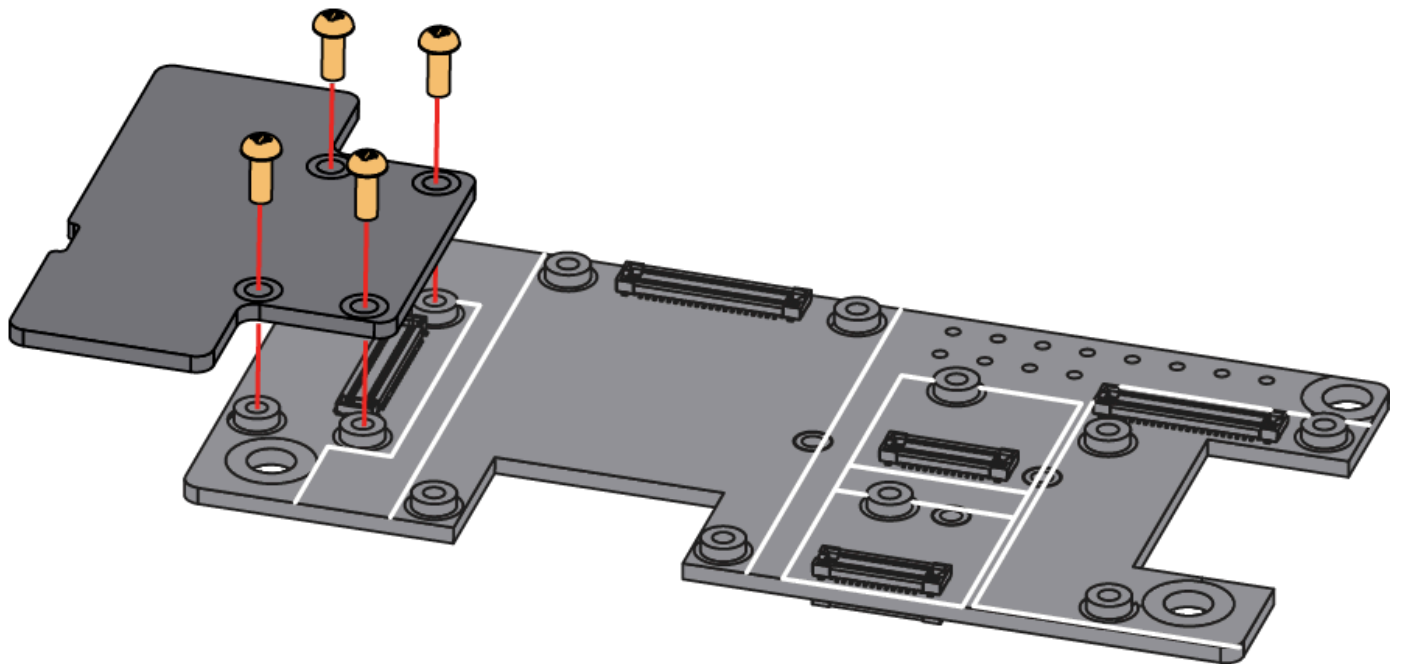


Figure 2: RAK19012 mounting mechanism on a WisBlock Base module

Hardware

The hardware specification is categorized into six parts. It discusses the interfacing, pinouts, and their corresponding functions and diagrams of the module. It also covers the electrical, mechanical, and environmental characteristics, including tabular data of the functionalities and standard values of the RAK19012 WisBlock LiPo Solar Power Slot Module.

Interfaces

The RAK19012 WisBlock LiPo Solar Power Slot Module provides the following interfaces:

- One WisBlock power module
- Two-pin battery interface
- Two-pin solar interface
- Three LEDs
- One reset button

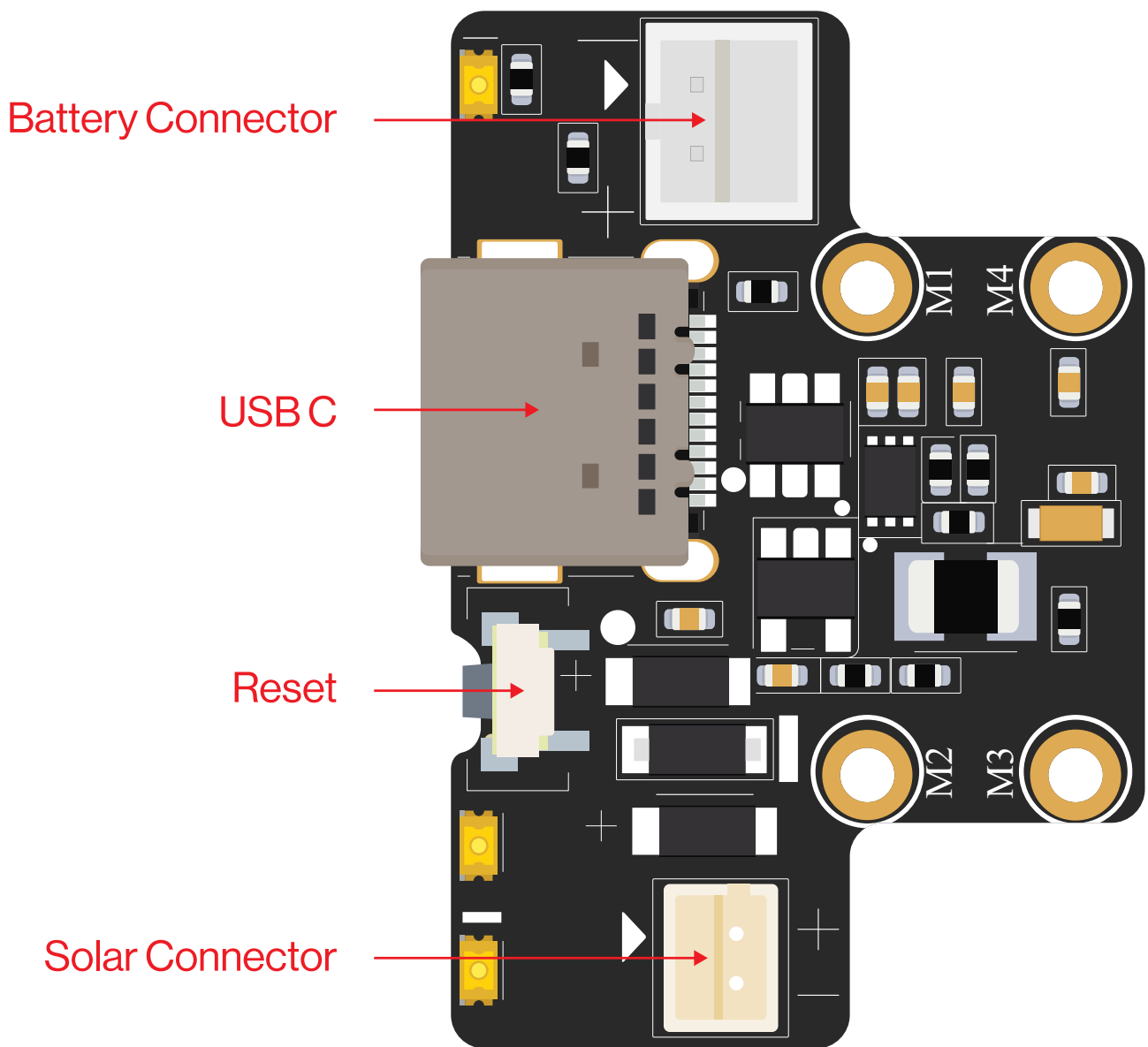


Figure 3: RAK19012 part labels

Battery and Solar Panel Connector

Figure 4 shows the polarity of battery and solar panel connectors.

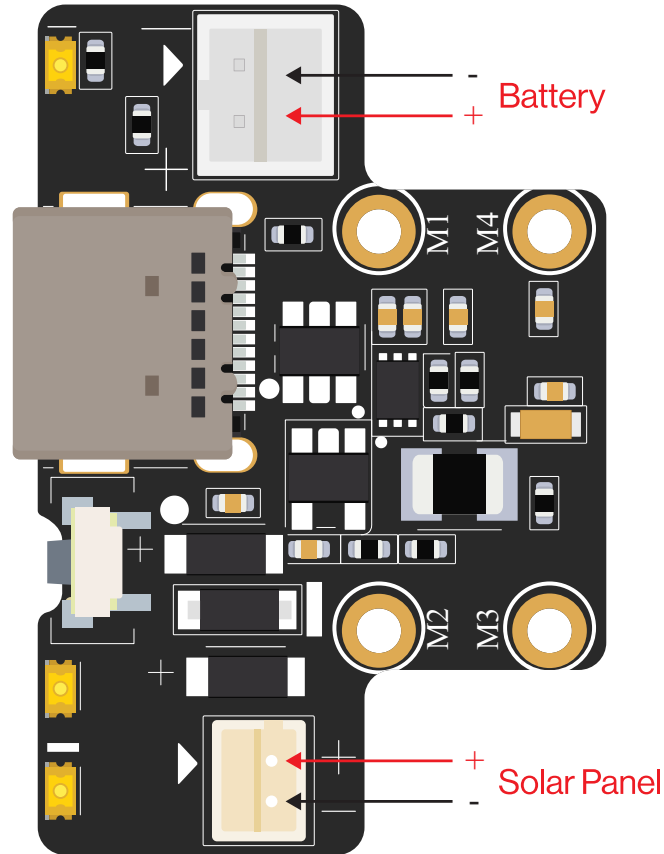


Figure 4: Battery and solar panel connector polarity

LEDs

Three LEDs indicate the operating status. Their functions are:

- **Red LED:** Connected to the charger chip to indicate charging status. It illuminates while the battery charges and dims or turns off when the battery is full.
- **Green LED:** Connected to the MCU module and controlled by user-defined MCU settings.

- **Blue LED:** Connected to the MCU module and controlled by user-defined MCU settings.

RESET Push Button

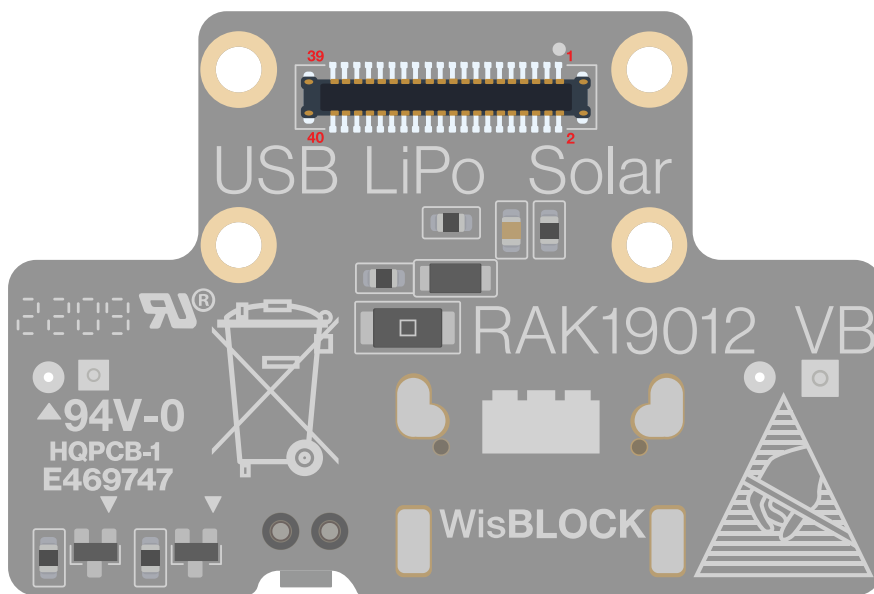
The Reset Push Button, as shown in [Figure 3](#), is connected to the MCU module. When pushed, it resets the MCU.

Pin Definition

The RAK19012 module has a 40-pin WisConnector that is compatible with the WisBlock Power Slot. The pin order of the connector and the pinout definition is shown in [Figure 5](#).

NOTE

VBAT, 3V3, RESET, LED1, LED2, ADC_VBAT, and GND are connected to the WisIO connector.



VBAT	2	1	VBAT
GND	4	3	GND
3v3	6	5	3v3
USB_N	8	7	USB_P
NC	10	9	VBUS
NC	12	11	NC
LED1	14	13	RESET
NC	16	15	LED2
NC	18	17	NC
NC	20	19	NC
NC	22	21	ADC_VBAT
NC	24	23	NC
NC	26	25	NC
NC	28	27	NC
NC	30	29	NC
NC	32	31	NC
NC	34	33	NC
NC	36	35	NC
CHRG	38	37	NC
GND	40	39	GND

Figure 5: RAK19012 pinout diagram

Electrical Characteristics

Absolute Maximum Ratings

The absolute maximum ratings of the device are shown in the table below. Stress ratings are for the functional operation of the device.

WARNING

1. If the stress rating exceeds the listed value, it may cause permanent damage to the device.
2. Exposure to maximum rated conditions may affect device reliability.

Ratings	Maximum Value	Unit
Battery voltage (VBAT)	-0.3 to 4.3	V
Solar panel voltage (VIN)	-0.3 to 5.5	V
IOs of WisBlock Connector	-0.3 to VDD+0.3	V

Battery Specification

The RAK19012 USB LiPo Solar Power Slot Module can be powered by a battery connected to the J4 connector. The battery's nominal operating voltage should be within the range shown in the following table.

Minimum	Typical	Maximum	Unit
3.3	3.7	4.3	V

A suitable Li-ion battery should have the following parameters, as shown in the table below:

Parameter	Value
Standard voltage	3.7 V
Charging voltage	4.2 V
Capacity	As required
Discharge current	At least 500 mA

NOTE

When using a solar panel, you can't use a non-rechargeable battery.

Mechanical Characteristic

Board Dimensions

The mechanical dimensions of the RAK19012 module are shown in **Figure 6** below.

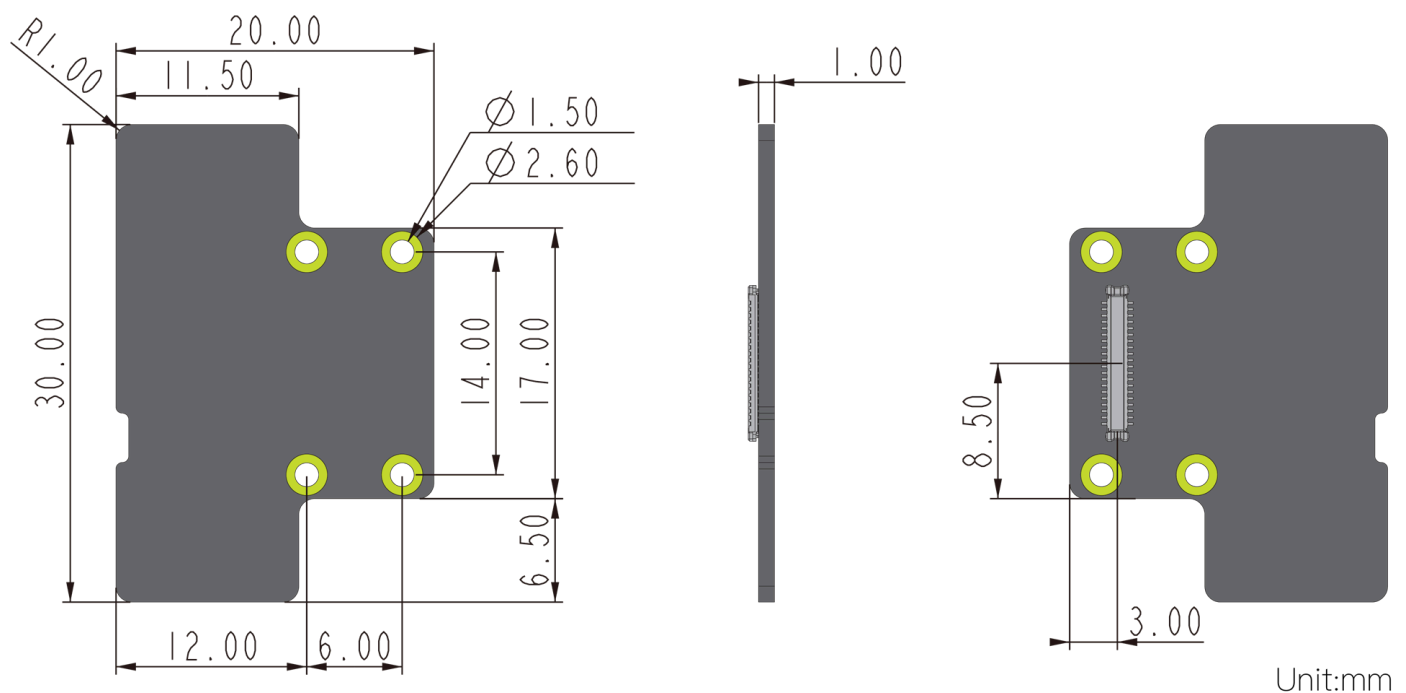


Figure 6: RAK19012 mechanical dimensions

WisConnector PCB Layout

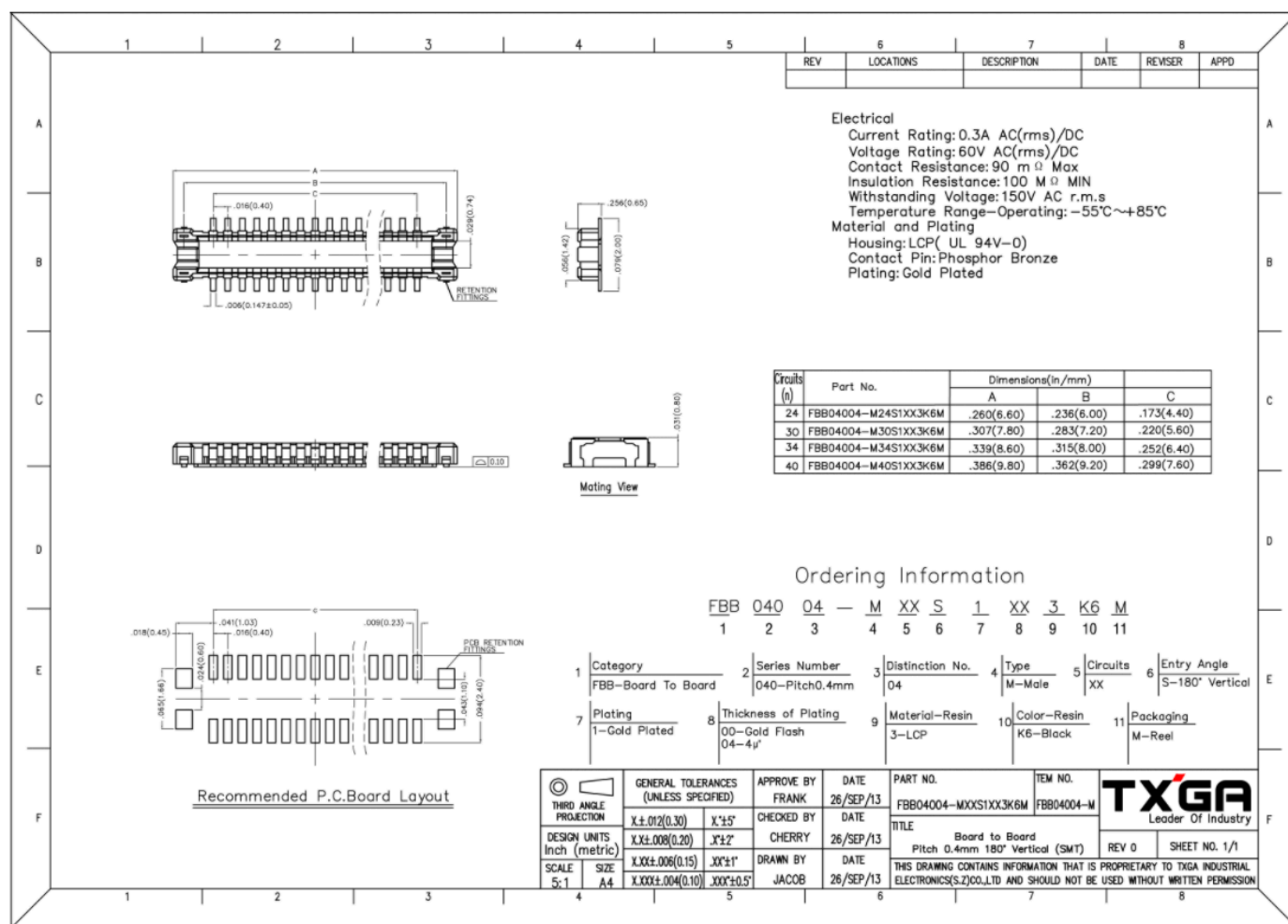


Figure 7: WisConnector PCB footprint and recommendations

Environmental Characteristics

The table below lists the operating and storage temperature requirements of the RAK19012:

Parameter	Minimum	Typical	Maximum
Operational temperature range	-35 °C	+25 °C	+75 °C
Extended temperature range	-40 °C	+25 °C	+80 °C
Storage temperature range	-40 °C	+25 °C	+80 °C

Schematic Diagram

Figure 8 shows the schematic of the RAK19012 USB LiPo Solar Power Slot Module.

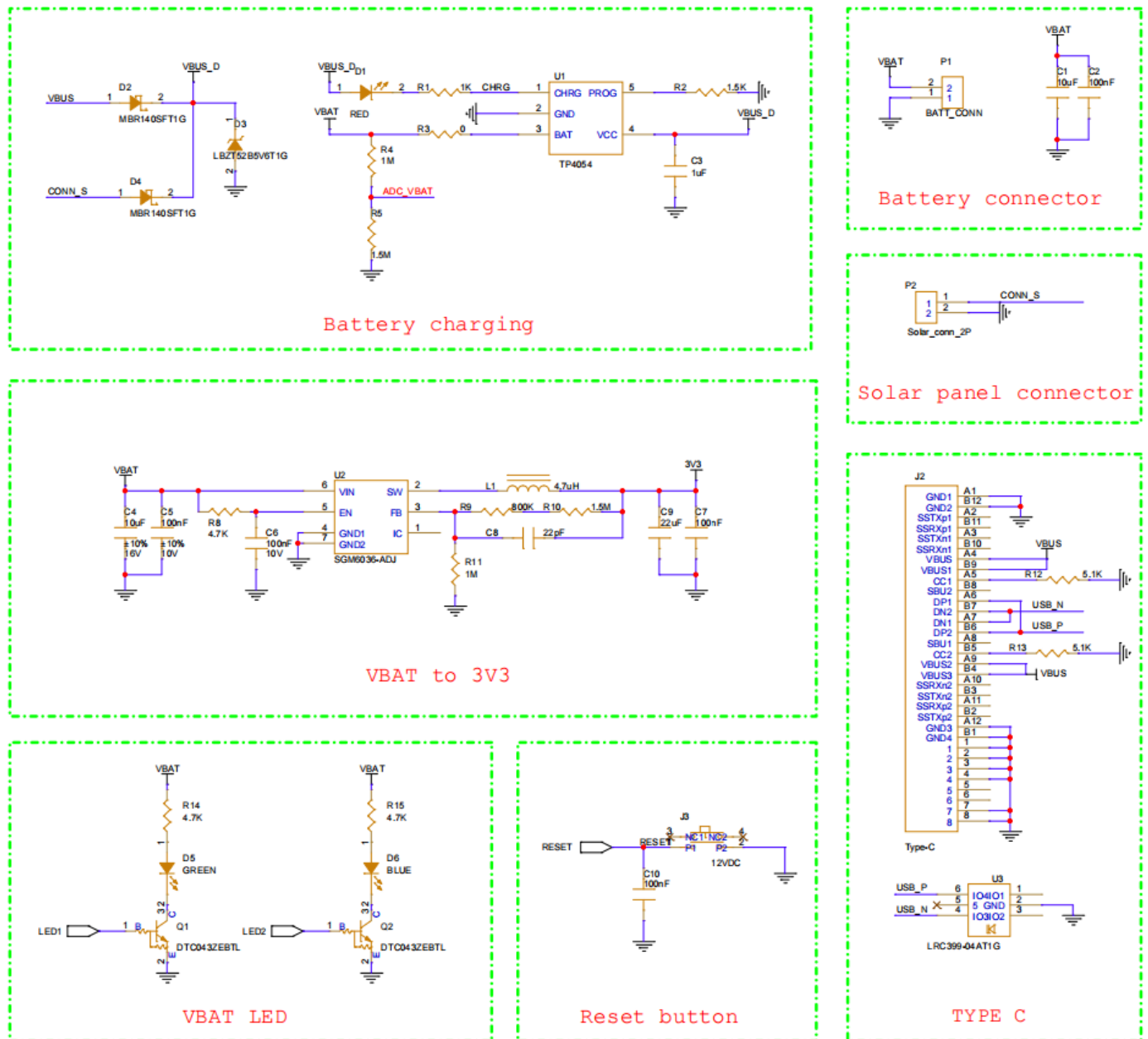


Figure 8: RAK19012 USB LiPo Solar Power Slot Module schematics